COM759: Knowledge Engineering & AI

15 Multiple Choice Questions: Semantic Web & Ontology Engineering

Week 5 Source Material Grounding Assessment for Master's Degree Students in Computer Science

Q1. How does the architectural and functional paradigm of Web 3.0 fundamentally differ from Web 1.0 ('read-only') and Web 2.0 ('read-write')?

A) Web 3.0 restricts content generation exclusively to centralized corporate institutions.

B) Web 3.0 combines the Semantic Web and Artificial Intelligence to enable machines to process data meaning and generate new information collaboratively.

C) Web 3.0 relies entirely on static HTML markup without URIs or client-side rendering.

D) Web 3.0 replaces machine-to-machine protocols with manual human tagging folksonomies.

Correct Answer: B) Web 3.0 combines the Semantic Web and Artificial Intelligence to enable machines to process data meaning and generate new information collaboratively.

Explanation: According to the Week 5 source material, Web 1.0 was a 'read-only' web, Web 2.0 introduced 'read-write' user-generated content, and Web 3.0 integrates the Semantic Web and Artificial Intelligence as cornerstones, allowing online applications to process data semantics, learn, and deliver new machine-interpreted information.

Q2. What is identified in the source material as the primary obstacle of the traditional 'Syntactic Web' regarding automated software agent execution?

A) Web content markup specifies presentation and hyperlinks, making semantic meaning accessible to humans but not easily processable by machines.

B) Computers lack the network bandwidth required to perform string-matching search queries.

C) Hypertext links do not support cross-domain page navigation.

D) Search engines operate with low recall and high precision across unstructured HTML documents.

Correct Answer: A) Web content markup specifies presentation and hyperlinks, making semantic meaning accessible to humans but not easily processable by machines.

Explanation: The source highlights that in the Syntactic Web, computers handle presentation while humans perform linking and interpretation. The core obstacle for automated agents is that web content meaning is not machine-accessible, resulting in search results with high recall but low precision.

Q3. Who coined the term Semantic Web in 2001, and how was its overarching vision defined in *Scientific American*?

A) Darcy DiNucci; as a platform for blog syndication and social media reviews.

B) Tim Berners-Lee (along with James Hendler and Ora Lassila); as an extension of the existing web in which information is given well-defined meaning to enable computers and people to work in cooperation.

C) Tim O'Reilly; as a folksonomy tagging network for photo sharing repositories.

D) Marvin Minsky; as a centralized frame database replacing distributed HTML documents.

Correct Answer: B) Tim Berners-Lee (along with James Hendler and Ora Lassila); as an extension of the existing web in which information is given well-defined meaning to enable computers and people to work in cooperation.

Explanation: Tim Berners-Lee (inventor of the WWW and W3C director), along with James Hendler and Ora Lassila (Scientific American, May 2001), coined and defined the Semantic Web as an extension of the current Web where information is given well-defined meaning to facilitate machine-human cooperation.

Q4. According to the widely cited definition by Thomas Gruber (1993) and Rudi Studer et al. (1998), what is an Ontology in computer science?

A) A relational database table storing raw numerical sensory measurements.

B) A formal, explicit specification of a shared conceptualisation.

C) A procedural search algorithm executed over unindexed web pages.

D) A non-symbolic neural network architecture for natural language parsing.

Correct Answer: B) A formal, explicit specification of a shared conceptualisation.

Explanation: The canonical definition established by Gruber (1993) and refined by Studer et al. (1998) defines an ontology as a 'formal, explicit specification of a shared conceptualisation', where 'formal' implies machine-readability with computational semantics, 'explicit' defines concept types and constraints, and 'shared' captures consensual domain knowledge.

Q5. How do Ontologies fundamentally differ from traditional Quantitative Numerical Models when describing domain knowledge?

A) Quantitative models capture qualitative descriptive data (e.g., 'cold', 'drunk'), while ontologies store precise numerical millimeter measurements.

B) Quantitative models process unambiguous numerical measurements, whereas ontologies capture qualitative, descriptive domain understandings using concepts, relations, and constraints.

C) Ontologies eliminate the need for class hierarchies and property value restrictions.

D) Quantitative models rely on formal description logic reasoners, whereas ontologies rely on statistical regression.

Correct Answer: B) Quantitative models process unambiguous numerical measurements, whereas ontologies capture qualitative, descriptive domain understandings using concepts, relations, and constraints.

Explanation: The source specifies that quantitative models measure the world using unambiguous numerical data (e.g., voltages, ages), whereas ontologies capture qualitative descriptions of domain understanding (concepts, attributes, eaten food types) to enable automated logical processing.

Q6. In the W3C Semantic Web architecture (Layer Cake), which standard layer is responsible for assigning unambiguous global names to resources, and which layer provides the data interchange model?

A) XML for global naming; SPARQL for data interchange.

B) URI/IRI for global naming; RDF (Resource Description Framework) for data interchange.

C) RDFS for global naming; RIF (Rule Interchange Format) for data interchange.

D) OWL for global naming; Crypto for data interchange.

Correct Answer: B) URI/IRI for global naming; RDF (Resource Description Framework) for data interchange.

Explanation: In the W3C Semantic Web Layer Cake, URI/IRI sits at the foundational layer to assign unambiguous global identifiers, while RDF (Resource Description Framework) sits above XML to provide the standard data interchange format for metadata and facts.

Q7. What are the four main structural components that constitute a formal ontology?

A) Tables, Primary Keys, Foreign Keys, and SQL Queries.

B) Classes (Concepts), Properties (Slots/Attributes), Relations, and Axioms.

C) Neural Nodes, Edge Weights, Activation Functions, and Epochs.

D) HTML Tags, CSS Stylesheets, JavaScript Scripts, and HTTP Headers.

Correct Answer: B) Classes (Concepts), Properties (Slots/Attributes), Relations, and Axioms.

Explanation: An ontology is structurally composed of Classes (conceptual entities/sets), Properties/Slots (attributes), Relations (relationships between concepts/properties), and Axioms (logical assertions and constraints joining concepts and relations).

Q8. In an ontology taxonomy, if Class C is declared a subclass of Class C' (rdfs:subClassOf), what inheritance behavior applies to slots and instances?

A) Class C overrides and deletes all properties defined in Class C'.

B) Every individual belonging to Class C is automatically an instance of Class C', and Class C inherits all slots (properties) from C'.

C) Class C' inherits all instances and properties from Class C, but not vice versa.

D) Subclasses can only inherit from a single superclass under OWL Description Logic rules.

Correct Answer: B) Every individual belonging to Class C is automatically an instance of Class C', and Class C inherits all slots (properties) from C'.

Explanation: Subclassing creates an IS-A relationship where every instance of subclass C is included in superclass C'. Furthermore, subclasses inherit all slots (properties) from their superclasses, including multiple superclasses (e.g., Port inheriting from both Dessert Wine and Red Wine).

Q9. Which of the following correctly pairs an ontology property type with its functional definition?

A) Intrinsic properties represent external object relationships; Extrinsic properties represent primitive values.

B) Simple properties (attributes) contain primitive values (strings, numbers); Complex properties link or point to other instances/objects.

C) Symmetric properties require inverse properties to be explicitly defined under different names.

D) Transitive properties restrict slot cardinality to exactly one instance.

Correct Answer: B) Simple properties (attributes) contain primitive values (strings, numbers); Complex properties link or point to other instances/objects.

Explanation: Simple properties (datatype properties/attributes) store primitive data types like integers or strings, whereas complex properties (object properties) represent binary relations that point to other individuals/instances in the domain.

Q10. In standard ontology visual modeling conventions (as used in Protégé and Knowledge Engineering literature), how are Classes and Individuals graphically represented?

A) Classes are represented as diamonds; Individuals are represented as circles/ovals.

B) Classes are represented as circles or ovals; Individuals are represented as diamonds.

C) Classes are represented as directed arrows; Individuals are represented as rectangles.

D) Both Classes and Individuals are represented as shaded square boxes.

Correct Answer: B) Classes are represented as circles or ovals; Individuals are represented as diamonds.

Explanation: In standard ontology diagrams, Classes (sets containing instances) are drawn as circles or ovals, while Individuals (specific domain objects like Matthew or Italy) are drawn as diamonds.

Q11. When developing a domain ontology hierarchy, what characterizes the Combination (Salient-first) development strategy?

A) Defining the most general root concepts first and subsequently specializing them down the taxonomy.

B) Defining the most specific leaf instances first and grouping them into general superclasses.

C) Defining the most prominent and salient domain concepts first, then generalizing and specializing them iteratively.

D) Automatically generating class trees from unstructured natural language text without human intervention.

Correct Answer: C) Defining the most prominent and salient domain concepts first, then generalizing and specializing them iteratively.

Explanation: The combination approach identifies the salient, central concepts of the domain first, and then expands the ontology by generalizing upward to superclasses and specializing downward to subclasses.

Q12. What is the correct sequence of steps in the standard 5-step ontology construction methodology?

A) 1. Define slots -> 2. Enumerate terms -> 3. Define constraints -> 4. Determine domain/scope -> 5. Define hierarchy.

B) 1. Determine domain and scope -> 2. Enumerate important terms -> 3. Define classes and class hierarchy -> 4. Define slots -> 5. Define slot constraints.

C) 1. Write SPARQL queries -> 2. Build Protégé UI -> 3. Instantiate individuals -> 4. Define axioms -> 5. Test reasoner.

D) 1. Define slot cardinality -> 2. Choose URI namespace -> 3. Create RDF triples -> 4. Build database schema -> 5. Run inference.

Correct Answer: B) 1. Determine domain and scope -> 2. Enumerate important terms -> 3. Define classes and class hierarchy -> 4. Define slots -> 5. Define slot constraints.

Explanation: The standard 5-step engineering process outlined in the source is: Step 1: Determine domain and scope (using competency questions); Step 2: Enumerate important terms; Step 3: Define classes and class hierarchy; Step 4: Define slots (properties); Step 5: Define slot constraints (cardinality, value types).

Q13. What key guideline governs the scope and granularity when engineering a domain ontology?

A) An ontology must contain every possible concept and property within the general domain regardless of application requirements.

B) An ontology should only include salient properties and concepts required by the intended application, avoiding over-specialization.

C) There is always exactly one mathematically correct way to model any real-world domain.

D) Classes must strictly correspond to verbs in natural language descriptions.

Correct Answer: B) An ontology should only include salient properties and concepts required by the intended application, avoiding over-specialization.

Explanation: The engineering guidelines emphasize that an ontology should NOT attempt to contain all possible information about a domain. Modellers should limit scope to what the application requires, including only salient properties. Moreover, there is no single 'correct' way to model a domain.

Q14. What automated reasoning task detects a logical error when an individual is asserted or inferred as an instance of two classes that are declared Disjoint?

A) Class Equivalence Check.

B) Consistency Checking.

C) Subclass Taxonomical Generalization.

D) Inverse Property Matching.

Correct Answer: B) Consistency Checking.

Explanation: Consistency Checking evaluates the logical validity of an ontology. If an individual x is inferred or asserted to belong to Class A and Class B, but A and B are declared disjoint, the reasoner flags an inconsistency (indicating a modeling or data error).

Q15. Which of the following is NOT listed in the source material as a primary motivation for adopting ontologies in Knowledge Engineering?

A) To share a common understanding of information structure among people and software agents.

B) To enable reuse of domain knowledge and separate domain knowledge from operational knowledge.

C) To replace hardware-level RAM memory with sub-symbolic vector storage.

D) To make domain assumptions explicit and enable automated reasoning for query answering.

Correct Answer: C) To replace hardware-level RAM memory with sub-symbolic vector storage.

Explanation: The key reasons for using ontologies are: sharing common information structure, enabling knowledge reuse, making domain assumptions explicit, separating domain knowledge from operational knowledge, and enabling automated reasoning. Hardware memory replacement is not an ontology function.